# Michael Rizzo Smith

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## Education

**Vanderbilt University** Nashville, TN
*Ph.D Astrophysics* *Aug 2023 - Exp. 2029*
*Advisor: Kelly Holley-Bocklemann*

**Stony Brook University** Stony Brook, NY
*Bachelor of Science, Physics and Astronomy & Planetary Science* *Aug 2017 - May 2021*

*Magna Cum Laude with Honors in Physics*

## Awards and Fellowships

**Award Recipient**, NSF Graduate Research Fellow — Jun 2025
**Award Recipient**, Vanderbilt Graduate School Travel Grant — July 2024
**Award Recipient**, McMinn Research Award — May 2024
**Award Recipient**, Vanderbilt Robert T. Lagemann Award — Apr 2024
**Departmental Honors**, Dept. of Physics — May 2021
**Award Recipient**, Stony Brook Campus Scholarship — Aug 2017 - May 2021

## Professional Work and Teaching

**Teaching Assistant**, ASTR 1010L: Intro to Astronomy Lab — Fall 2023 - Spring 2025
**Co-Coordinator**, Vanderbilt's Graduate Student Astrophysics Journal Club — Aug 2024 - present
**Code Captain**, LISA, Disc-IMRI Astrophysics Working Group — Dec 2023 - present
**Member**, Establishing Multimessenger astronomy Inclusive Training (EMIT) — Sep 2023 - present
**Data Analyst**, The Ohio State University — Jun 2021 - July 2023

## Presentations

**Invited Speaker**, EMIT Outreach — Jan 2025
*Presented to a local public high school on computational astrophysics, use of simulations in science, and different pathways to academia.*

**Talk**, Vanderbilt's Astrophysics Journal Research Club — Jan 2025
*The dynamics and electromagnetic signatures of accretion in unequal mass binary black hole inspirals*

**Talk**, The 15th International LISA Symposium — July 2024
*Presenting on behalf of the Disc-IMRI Code Comparison LISA Astrophysics Working Group*

**Invited Speaker**, Valor Collegiate Academy — Mar 2024
*Invited to speak to first-year class of a local public high school on a career in Astrophysics and my personal path.*

**Talk**, Vanderbilt's Astrophysics Journal Research Club — Feb 2024
*Astrophysical Disk Modeling With Hydrodynamic Simulations*

## PUBLICATIONS

**Accepted or submitted to peer-reviewed scientific journals**

***(1 first-author, 3 co-author, 102 citations)***

1. J. M. M. Neustadt, C. S. Kochanek, and **Rizzo Smith, M.** *Constraints on pre-SN outbursts from the progenitor of SN 2023ixf using the large binocular telescope*. In: MNRAS 527.3 (Jan. 2024), pp. 5366–5373. arXiv: 2306.06162 `[astro-ph.HE]`
2. Matthew Kenworthy, [20 authors], and **Rizzo Smith, Michael**. *A planetary collision afterglow and transit of the resultant debris cloud*. In: Nature 622.7982 (Oct. 2023), pp. 251–254. arXiv: 2310.08360 `[astro-ph.EP]`
3. **Rizzo Smith, M.**, C. S. Kochanek, and J. M. M. Neustadt. *The late time optical evolution of twelve core-collapse supernovae: detection of normal stellar winds*. In: MNRAS 523.1 (July 2023), pp. 1474–1495. arXiv: 2212.09763 `[astro-ph.HE]`
4. A. Kawash, [11 authors], **Rizzo Smith, M.**, T. W. -S. Holoien, J. L. Prieto, and T. A. Thompson. *The Galactic Nova Rate: Estimates from the ASAS-SN and Gaia Surveys*. In: The Astrophysical Journal 937.2, 64 (Oct. 2022), p. 64. arXiv: 2206.14132

**Non-Referred Publications**

1. M. Rizzo Smith and ASASSN-Team. *An Update on ASASSN-21qj: A Rapidly Fading, Sun-Like Star; Back With a Vengeance*. In: The Astronomer's Telegram 15531 (July 2022), p. 1
2. M. Rizzo Smith and ASASSN-Team. *ASASSN-22el: A Deep Eclipse Event*. In: The Astronomer's Telegram 15308 (Apr. 2022), p. 1
3. M. Rizzo Smith and ASASSN-Team. *ASASSN-21sa: A Deep Dimming Event*. In: The Astronomer's Telegram 14937 (Sept. 2021), p. 1
4. M. Rizzo Smith and ASASSN-Team. *ASASSN-21qj: A Rapidly Fading, Sun-Like Star*. In: The Astronomer's Telegram 14879 (Aug. 2021), p. 1
5. M. Rizzo Smith and ASASSN-Team. *ASASSN-21nn: An Unusual Dimming Event in a Red Giant Star*. In: The Astronomer's Telegram 14803 (July 2021), p. 1
6. M. Rizzo Smith and ASASSN-Team. *ASASSN-21ml: Discovery of an Extreme (Delta g $>$ 10 mag) L-Dwarf Flare*. In: The Astronomer's Telegram 14778 (July 2021), p. 1